

Movie Rental Data Warehouse Design Report

High-Level Data Warehouse Design from an OLTP Movie Rental System

Course: Data Warehousing / Data Architecture

Assignment Type: Individual or Group

Main Topic: From OLTP Schema to Dimensional Model and ETL Design

Prepared Report

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This report proposes a dimensional Data Warehouse design for a movie rental business, including business questions, fact tables, dimension tables, a dimensional model diagram, ETL design, and data quality rules.

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1. Introduction

The purpose of this assignment is to design a high-level Data Warehouse for a movie rental business based on an existing OLTP database schema. The OLTP system supports daily operational activities such as registering customers, managing films, storing inventory, recording rentals, processing payments, managing staff, and maintaining store and location information.

An OLTP system is designed mainly for transaction processing. It focuses on storing daily business operations accurately and efficiently. For example, when a customer rents a film, returns a film, or makes a payment, the OLTP system records these transactions. However, OLTP systems are not optimized for analytical reporting because the data is usually normalized and spread across many related tables.

A Data Warehouse is designed for analysis and decision-making. Instead of focusing only on daily transactions, it organizes data in a way that helps managers understand business performance. For a movie rental business, the Data Warehouse can support analysis of rental activity, revenue trends, customer behavior, film popularity, store performance, staff performance, and location-based trends.

The goal of this assignment is not to copy the OLTP schema directly. Instead, the goal is to reorganize the operational data into a dimensional model that is easier to query and understand. The proposed design uses fact tables to store measurable business events and dimension tables to describe those events from different perspectives such as time, customer, film, store, staff, and location.

Assumption: The design is based on the standard Movie Rental / Sakila-style OLTP schema referenced by the assignment, using tables such as rental, payment, customer, film, inventory, store, staff, address, city, country, category, film_category, language, actor, and film_actor.

2. Business Questions

Before designing the Data Warehouse, it is important to identify the analytical questions that business managers want to answer. These questions help determine the required fact tables, dimensions, measures, and relationships.

Business Question	Area of Analysis	Importance for Decision-Making
Which films are rented most frequently?	Film popularity	Helps management identify the most demanded films and make better inventory decisions.
Which films generate the highest revenue?	Revenue analysis	Helps identify the most profitable films, not only the most rented ones.
Which film categories are most popular?	Category analysis	Helps the business understand customer preferences and invest in popular categories.
Which stores generate the highest number of rentals?	Store performance	Helps compare store activity and identify high-performing branches.
Which stores generate the highest revenue?	Store revenue	Helps management evaluate store profitability.
Which customers rent the most films?	Customer behavior	Helps identify loyal and active customers for rewards or marketing campaigns.
Which customers generate the highest revenue?	Customer value analysis	Helps identify valuable customers who contribute most to revenue.
How does rental activity change over time?	Time-based rental trends	Helps detect seasonal changes, peak periods, and low-demand periods.
How does revenue change by month, quarter, or year?	Time-based revenue trends	Helps management monitor financial performance over time.
Which staff members process the highest number of rentals?	Staff performance	Helps evaluate employee activity and workload.
Which staff members process the highest payment amounts?	Staff revenue contribution	Helps analyze staff involvement in payment processing.
Which cities or countries generate the highest customer activity?	Location analysis	Helps understand geographic demand and customer distribution.
What is the average rental duration for films or categories?	Rental duration analysis	Helps understand how long customers usually keep films.
Which films are returned late most often?	Late return analysis	Helps identify films or customers associated with late returns.
How does store performance differ by location?	Store/location comparison	Helps management make location-based business decisions.

These business questions guide the dimensional model design. Rental-related questions require a rental fact table, revenue-related questions require a payment fact table, and film availability questions require an inventory snapshot fact table. Dimensions such as Date, Customer, Film, Store, Staff, Location, Category, Actor, and Language are needed to analyze the facts from different perspectives.

3. Dimensional Model Design

The proposed Data Warehouse uses dimensional modeling. Dimensional modeling organizes data into fact tables and dimension tables. Fact tables store measurable business events, while dimension tables store descriptive information used for filtering, grouping, and reporting.

The main design style is a Star Schema because it is simple, easy to understand, and efficient for reporting. However, a small hybrid approach is used for many-to-many relationships, especially between films and categories, and between films and actors. These relationships are handled using bridge tables.

3.1 Selected Business Processes

Business Process	Description	Why It Is Important for Analysis
Rental Transactions	Represents each time a customer rents a film.	Needed to analyze rental counts, rental duration, late returns, and film popularity.
Payment Transactions	Represents each payment made by a customer.	Needed to analyze revenue, payment amounts, customer value, and store profitability.
Film Inventory / Availability	Represents the availability of films in stores.	Needed to analyze inventory levels, available films, and rented copies.
Customer Rental Behavior	Represents how customers interact with the rental business.	Helps identify active customers, valuable customers, and customer trends.
Store Performance	Represents rental and revenue activity by store.	Helps compare branches and evaluate business performance by location.
Staff Processing Activity	Represents rentals and payments processed by staff.	Helps evaluate staff workload and performance.

3.2 Fact Tables

3.2.1 Fact_Rental

Business Process: The Fact_Rental table represents film rental transactions.

Grain: One row represents one rental transaction.

Purpose: This fact table is used to analyze rental activity, rental frequency, rental duration, late returns, film popularity, customer rental behavior, store performance, and staff activity.

Foreign Key	References
rental_date_key	Dim_Date
return_date_key	Dim_Date
customer_key	Dim_Customer
film_key	Dim_Film
store_key	Dim_Store
staff_key	Dim_Staff

Measure	Description
rental_count	Usually equals 1 for each rental transaction.
rental_duration_days	Number of days between rental date and return date.
late_return_flag	Indicates whether the film was returned late.
late_days	Number of late days if the film was returned after the expected rental duration.

3.2.2 Fact_Payment

Business Process: The Fact_Payment table represents payment transactions.

Grain: One row represents one payment transaction.

Purpose: This fact table is used to analyze revenue, payment amounts, customer value, store revenue, staff performance, and revenue trends over time.

Foreign Key	References
payment_date_key	Dim_Date
customer_key	Dim_Customer
staff_key	Dim_Staff
store_key	Dim_Store
film_key	Dim_Film, if derivable through rental/inventory

Measure	Description
payment_amount	Amount paid by the customer.
payment_count	Usually equals 1 for each payment transaction.
revenue	Same as payment amount for revenue reporting.

3.2.3 Fact_Inventory_Snapshot

Business Process: The Fact_Inventory_Snapshot table represents film inventory and availability by store and date.

Grain: One row represents one film in one store on one snapshot date.

Purpose: This fact table is used to analyze film availability, rented copies, available copies, and inventory distribution across stores.

Foreign Key	References
snapshot_date_key	Dim_Date
film_key	Dim_Film
store_key	Dim_Store

Measure	Description
inventory_count	Total number of copies of a film in a store.
available_count	Number of available copies.
rented_count	Number of currently rented copies.

Fact Tables Summary

Fact Table	Business Process	Grain	Main Measures	Related Dimensions
Fact_Rental	Rental transactions	One row per rental transaction	rental_count, rental_duration_days, late_return_flag, late_days	Date, Customer, Film, Store, Staff
Fact_Payment	Payment transactions	One row per payment transaction	payment_amount, payment_count, revenue	Date, Customer, Staff, Store, Film
Fact_Inventory_Snapshot	Inventory availability	One row per film per store per snapshot date	inventory_count, available_count, rented_count	Date, Film, Store

Fact Table Columns

Fact Table	Column Name	Type of Column	Description
Fact_Rental	rental_fact_key	Surrogate Key	Unique key for each rental fact row.
Fact_Rental	rental_id	Degenerate Dimension	Original rental transaction ID from OLTP.
Fact_Rental	rental_date_key	Foreign Key	Links to rental date in Dim_Date.
Fact_Rental	return_date_key	Foreign Key	Links to return date in Dim_Date.
Fact_Rental	customer_key	Foreign Key	Links to Dim_Customer.
Fact_Rental	film_key	Foreign Key	Links to Dim_Film.
Fact_Rental	store_key	Foreign Key	Links to Dim_Store.
Fact_Rental	staff_key	Foreign Key	Links to Dim_Staff.
Fact_Rental	rental_count	Measure	Count of rental transaction.
Fact_Rental	rental_duration_days	Measure	Duration between rental and return.
Fact_Rental	late_return_flag	Measure	Indicates if rental was returned late.
Fact_Rental	late_days	Measure	Number of days late.
Fact_Payment	payment_fact_key	Surrogate Key	Unique key for each payment fact row.
Fact_Payment	payment_id	Degenerate Dimension	Original payment transaction ID from

			OLTP.
Fact_Payment	payment_date_key	Foreign Key	Links to payment date in Dim_Date.
Fact_Payment	customer_key	Foreign Key	Links to Dim_Customer.
Fact_Payment	staff_key	Foreign Key	Links to Dim_Staff.
Fact_Payment	store_key	Foreign Key	Links to Dim_Store.
Fact_Payment	film_key	Foreign Key	Links to Dim_Film if derived through rental.
Fact_Payment	rental_id	Degenerate Dimension	Original rental reference.
Fact_Payment	payment_amount	Measure	Amount paid by customer.
Fact_Payment	payment_count	Measure	Count of payment transaction.
Fact_Inventory_Snapshot	snapshot_fact_key	Surrogate Key	Unique key for each inventory snapshot row.
Fact_Inventory_Snapshot	snapshot_date_key	Foreign Key	Links to Dim_Date.
Fact_Inventory_Snapshot	film_key	Foreign Key	Links to Dim_Film.
Fact_Inventory_Snapshot	store_key	Foreign Key	Links to Dim_Store.
Fact_Inventory_Snapshot	inventory_count	Measure	Total copies in inventory.
Fact_Inventory_Snapshot	available_count	Measure	Available copies.
Fact_Inventory_Snapshot	rented_count	Measure	Rented copies.

3.3 Dimension Tables

Dimension Table	Main Attributes	Source OLTP Tables	Purpose
Dim_Date	date_key, full_date, day, month, month_name, quarter, year, day_of_week, is_weekend	Generated from rental, return, payment, and snapshot dates	Supports time-based analysis such as monthly, quarterly, and yearly trends.
Dim_Customer	customer_key, customer_id, customer_full_name, email, active_status, create_date, location_key	customer, address, city, country	Supports customer behavior and customer value analysis.
Dim_Film	film_key, film_id, title, description, release_year, rental_duration, rental_rate, length, replacement_cost, rating, special_features, language_key	film, language	Supports film popularity, revenue, duration, and rating analysis.
Dim_Category	category_key, category_id, category_name	category, film_category	Supports analysis by film category.
Dim_Store	store_key, store_id, manager_staff_id, location_key	store, staff, address, city, country	Supports store performance and store revenue analysis.
Dim_Staff	staff_key, staff_id, staff_full_name, email, active_status, username, store_key, location_key	staff, store, address, city, country	Supports staff performance analysis.
Dim_Location	location_key, address_id, address, district, city, country, postal_code, phone	address, city, country	Supports city, country, and location-based analysis.
Dim_Language	language_key, language_id, language_name	language	Supports analysis by film language.
Dim_Actor	actor_key, actor_id, actor_full_name	actor, film_actor	Supports film and actor-based analysis.
Bridge_Film_Category	film_key, category_key, allocation_weight	film_category	Handles many-to-many relationship between films and categories.
Bridge_Film_Actor	film_key, actor_key, allocation_weight	film_actor	Handles many-to-many relationship between films and actors.

Shared Dimensions

Shared Dimension	Used By
Dim_Date	Fact_Rental, Fact_Payment, Fact_Inventory_Snapshot
Dim_Customer	Fact_Rental, Fact_Payment
Dim_Film	Fact_Rental, Fact_Payment, Fact_Inventory_Snapshot
Dim_Store	Fact_Rental, Fact_Payment, Fact_Inventory_Snapshot
Dim_Staff	Fact_Rental, Fact_Payment
Dim_Location	Customer, Store, and Staff analysis

Slowly Changing Dimensions

Some dimension attributes may change over time. For example, a customer may move to a new address, a staff member may change store, or a film attribute may be updated.

- SCD Type 1 can be used for simple corrections where historical tracking is not required.
- SCD Type 2 can be used when the business wants to preserve historical changes, such as customer location history or staff store assignment history.

4. Dimensional Model Diagram

The following figure shows the proposed dimensional model for the Movie Rental Data Warehouse.

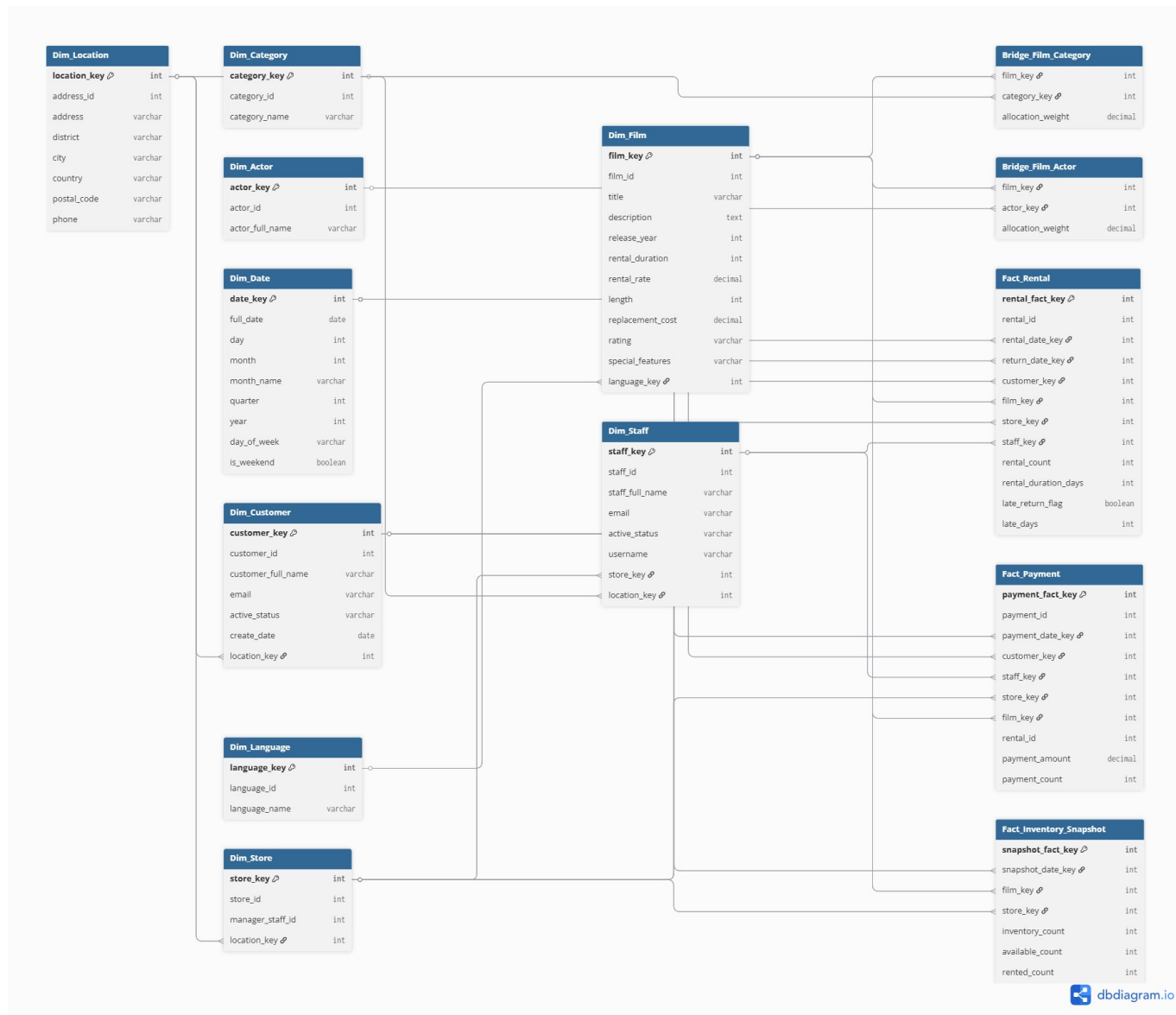


Figure 1: Proposed Movie Rental Data Warehouse Dimensional Model

The diagram shows the proposed dimensional model for the Movie Rental Data Warehouse. The design uses a star schema with a hybrid approach for many-to-many relationships. Fact_Rental, Fact_Payment, and Fact_Inventory_Snapshot represent the main business processes. These fact tables are connected to shared dimensions such as Dim_Date, Dim_Customer, Dim_Film, Dim_Store, Dim_Staff, and Dim_Location.

Bridge tables are used to handle many-to-many relationships between films and categories, and between films and actors. This prevents duplicated fact records and allows accurate analysis of film categories and actors.

5. ETL Design

ETL stands for Extract, Transform, and Load. It describes how data is moved from the OLTP system into the Data Warehouse.

5.1 Extract

In the Extract phase, data is selected from the OLTP source tables. The selected tables are used to build the dimensions and fact tables in the Data Warehouse.

Source OLTP Table	Data Extracted	Purpose in Data Warehouse
rental	rental_id, rental_date, inventory_id, customer_id, return_date, staff_id	Used to build Fact_Rental and calculate rental duration and late returns.
payment	payment_id, customer_id, staff_id, rental_id, amount, payment_date	Used to build Fact_Payment and analyze revenue.
customer	customer_id, store_id, first_name, last_name, email, address_id, active, create_date	Used to build Dim_Customer.
film	film_id, title, description, release_year, language_id, rental_duration, rental_rate, length, replacement_cost, rating, special_features	Used to build Dim_Film.
inventory	inventory_id, film_id, store_id	Used to link rentals to films and stores, and to build inventory analysis.
store	store_id, manager_staff_id, address_id	Used to build Dim_Store and analyze store performance.
staff	staff_id, first_name, last_name, address_id, email, store_id, active, username	Used to build Dim_Staff.
address	address_id, address, district, city_id, postal_code, phone	Used to build Dim_Location.
city	city_id, city, country_id	Used to build Dim_Location.
country	country_id, country	Used to build Dim_Location.
category	category_id, name	Used to build Dim_Category.
film_category	film_id, category_id	Used to build Bridge_Film_Category.
language	language_id, name	Used to build Dim_Language.
actor	actor_id, first_name, last_name	Used to build Dim_Actor.
film_actor	actor_id, film_id	Used to build Bridge_Film_Actor.

5.2 Transform

In the Transform phase, the extracted OLTP data is cleaned, joined, standardized, and prepared for loading into the Data Warehouse.

5.2.1 Joining Related Tables

- customer is joined with address, city, and country to create customer location details.
- store is joined with address, city, and country to create store location details.
- staff is joined with store and location tables to identify staff store and location information.
- film is joined with language to include language information in film analysis.
- film_category is joined with category to support category analysis.
- film_actor is joined with actor to support actor analysis.
- rental is joined with inventory to identify the rented film and store.
- payment is joined with rental if film-level payment analysis is needed.

5.2.2 Creating Full Names

- customer.first_name + customer.last_name becomes customer_full_name.
- staff.first_name + staff.last_name becomes staff_full_name.
- actor.first_name + actor.last_name becomes actor_full_name.

5.2.3 Cleaning Data

- Remove duplicate records.
- Handle missing values.
- Validate dates.

- Check that payment amounts are not negative.
- Standardize text values such as city, country, category, and language names.
- Replace missing location values with Unknown.

5.2.4 Creating Surrogate Keys

- The Data Warehouse should use surrogate keys instead of relying only on OLTP operational IDs.
- Examples include customer_key for Dim_Customer, film_key for Dim_Film, store_key for Dim_Store, and staff_key for Dim_Staff.
- Fact tables reference these surrogate keys.

5.2.5 Creating Date Keys

- rental_date_key from rental_date.
- return_date_key from return_date.
- payment_date_key from payment_date.
- snapshot_date_key for inventory snapshots.
- Dim_Date stores day, month, quarter, year, day of week, and weekend flag.

5.2.6 Calculating Measures

Measure	Calculation
rental_count	Set to 1 for each rental transaction.
payment_count	Set to 1 for each payment transaction.
rental_duration_days	return_date - rental_date.
payment_amount	Taken from the payment amount field.
revenue	Same as payment amount for reporting.
late_return_flag	True if actual rental duration is greater than expected rental duration.
late_days	actual rental duration - expected rental duration, only when positive.
inventory_count	Count of inventory copies by film and store.
available_count	Count of copies not currently rented.
rented_count	Count of copies currently rented.

5.2.7 Handling Null Return Dates

- If return_date is null, it means the film has not been returned yet.
- The rental should be marked as open or active.
- Final rental duration should not be treated as completed.
- The ETL process should not fail.
- A special date key such as Unknown or Not Returned may be used for return_date_key.

5.2.8 Handling Many-to-Many Relationships

Many-to-many relationships should not be stored directly inside fact tables because they may duplicate measures. Bridge_Film_Category handles the relationship between films and categories, while Bridge_Film_Actor handles the relationship between films and actors. These bridge tables allow reporting by category and actor without incorrectly duplicating rental or payment facts.

5.2.9 Handling Historical Changes

Slowly Changing Dimensions may be used for attributes that change over time, such as customer address changes, staff store changes, store manager changes, and film attribute changes. SCD Type 1 can overwrite old values when historical tracking is not required. SCD Type 2 can preserve history by creating a new dimension row with effective dates.

5.3 Load

In the Load phase, transformed data is inserted into the Data Warehouse tables. Dimension tables should be loaded before fact tables because fact tables depend on dimension surrogate keys. For example, Fact_Rental cannot be loaded until the related customer, film, store, staff, and date records exist in their dimension tables.

Suggested Loading Order

1. Dim_Date
2. Dim_Location
3. Dim_Customer
4. Dim_Store
5. Dim_Staff
6. Dim_Language
7. Dim_Film
8. Dim_Category
9. Dim_Actor
10. Bridge_Film_Category
11. Bridge_Film_Actor
12. Fact_Rental
13. Fact_Payment
14. Fact_Inventory_Snapshot

Loading Process

- Generate surrogate keys for all dimension records.
- Load dimension tables first.
- Maintain a lookup mapping between OLTP IDs and Data Warehouse surrogate keys.
- Load bridge tables after related dimensions are available.
- Load fact tables using dimension surrogate keys.
- Insert new records into dimensions and facts.
- Handle changed records using SCD Type 1 or SCD Type 2.
- Apply data quality checks before and after loading.
- Log rejected or invalid records for review.

6. Data Quality Considerations

Data quality checks are important because incorrect or inconsistent data can lead to misleading reports and poor business decisions.

Data Quality Rule	Reason	Action if Rule Fails
payment_amount must be greater than or equal to 0.	Negative payments can produce incorrect revenue reports.	Reject or log the record for review.
rental_date must not be null.	Every rental transaction must have a start date.	Reject the rental record or send it to an error table.
return_date must be after rental_date when it exists.	Prevents invalid rental duration calculations.	Log the record and exclude it from duration analysis until corrected.
customer_id must exist in the customer table.	Prevents facts from referencing missing customers.	Reject or assign to an Unknown Customer record.
film_id must exist through inventory and film.	Ensures each rental is linked to a valid film.	Reject or log the record.
staff_id must exist in the staff table.	Ensures rentals and payments are linked to valid staff.	Reject or assign to Unknown Staff.
store_id must exist in the store table.	Ensures store analysis is accurate.	Reject or assign to Unknown Store.
rental_id must be unique in Fact_Rental.	Prevents duplicate rental counts.	Remove duplicates or keep the most recent valid record.
payment_id must be unique in Fact_Payment.	Prevents duplicate revenue calculations.	Remove duplicates or log duplicates for review.
Missing city or country should be handled as Unknown.	Allows location analysis without breaking reports.	Replace missing values with Unknown.
Invalid dates should be rejected or logged.	Prevents incorrect time-based reporting.	Send invalid records to an error log.
Late return calculation must be validated.	Ensures late return reports are accurate.	Recalculate or log inconsistent records.
Null return_date should not break ETL.	Some rentals may still be active.	Mark rental as open or use an Unknown return date key.
Category and actor relationships should not duplicate fact rows.	Prevents overcounting rentals or revenue.	Use bridge tables instead of duplicating fact records.
All fact foreign keys must match valid dimension surrogate keys.	Maintains referential integrity.	Reject or assign to Unknown dimension rows.

7. Conclusion

This report proposed a high-level Data Warehouse design for a movie rental business. The design converts the operational OLTP structure into a dimensional model that is more suitable for analytical reporting and decision-making.

The proposed model includes three main fact tables: Fact_Rental, Fact_Payment, and Fact_Inventory_Snapshot. These fact tables support analysis of rental activity, revenue, inventory availability, customer behavior, store performance, and staff activity. The design also includes several dimension tables, such as Dim_Date, Dim_Customer, Dim_Film, Dim_Store, Dim_Staff, Dim_Location, Dim_Category, Dim_Language, and Dim_Actor.

A star schema is used as the main design because it is simple and efficient for reporting. A hybrid approach is used for many-to-many relationships through Bridge_Film_Category and Bridge_Film_Actor.

The ETL process explains how data is extracted from the OLTP source tables, transformed into clean and meaningful analytical structures, and loaded into the Data Warehouse. Data quality rules are also defined to ensure accurate reporting and reliable decision-making.

Overall, the proposed Data Warehouse helps business managers answer important questions about revenue, rentals, films, customers, stores, staff, locations, and trends over time. This makes the design more useful for analysis than the original OLTP schema.

Appendix A. GitHub README.md Content

Movie Rental Data Warehouse Design

Project Title

Movie Rental Data Warehouse Design

Course

Data Warehousing / Data Architecture

Assignment Topic

High-Level Data Warehouse Design from an OLTP Movie Rental System

Description

This project presents a high-level Data Warehouse design for a movie rental business. The goal is to transform an OLTP movie rental schema into a dimensional model suitable for analytical reporting and decision-making.

The design focuses on business questions, fact tables, dimension tables, dimensional modeling, ETL planning, data quality rules, and a dimensional model diagram.

Main Components

- Business questions
- Dimensional model design
- Fact tables
- Dimension tables
- Bridge tables
- ETL design
- Data quality considerations
- Dimensional model diagram

Fact Tables

- Fact_Rental

- Fact_Payment
- Fact_Inventory_Snapshot

Dimension Tables

- Dim_Date
- Dim_Customer
- Dim_Film
- Dim_Category
- Dim_Store
- Dim_Staff
- Dim_Location
- Dim_Language
- Dim_Actor

Bridge Tables

- Bridge_Film_Category
- Bridge_Film_Actor

ETL Summary

The ETL process extracts data from OLTP tables such as rental, payment, customer, film, inventory, store, staff, address, city, country, category, film_category, language, actor, and film_actor.

The transformation phase includes joining related tables, creating full names, cleaning data, generating surrogate keys, creating date keys, calculating measures, handling null return dates, and managing many-to-many relationships.

The loading phase loads dimension tables before fact tables and ensures that all fact tables reference valid dimension surrogate keys.

Repository Contents

- Final report
- Dimensional model diagram
- OLTP schema or schema notes
- Assignment file
- Supporting documentation

Appendix B. Suggested GitHub Repository Structure

movie-rental-data-warehouse/

```

├── README.md
├── report/
│   ├── Movie_Rental_Data_Warehouse_Report.docx
│   └── Movie_Rental_Data_Warehouse_Report.pdf
├── schema/
│   ├── oltp_schema.sql
│   └── schema_notes.md
├── diagrams/
│   └── dimensional_model_diagram.png
└── docs/

```

Appendix C. Submission Note to Instructor

Dear Instructor,

Please find below the GitHub repository link for my Data Warehousing assignment:

[GitHub Link]

The repository includes the final report, the dimensional model diagram, the OLTP schema or schema notes, and the supporting documentation.

Best regards,

Izz Aldeen Mansour

References

- [1] Course Assignment PDF, Data Warehousing Assignment - Movie Rental Data Warehouse Design, Data Warehousing / Data Architecture course.
- [2] Oracle MySQL Documentation, Sakila Sample Database. Available at: <https://dev.mysql.com/doc/sakila/en/>
- [3] Kimball Group, Dimensional Modeling Techniques. Available at: <https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/kimball-techniques/dimensional-modeling-techniques/>
- [4] dbdiagram.io, Database Relationship Diagrams Design Tool. Available at: <https://dbdiagram.io/>